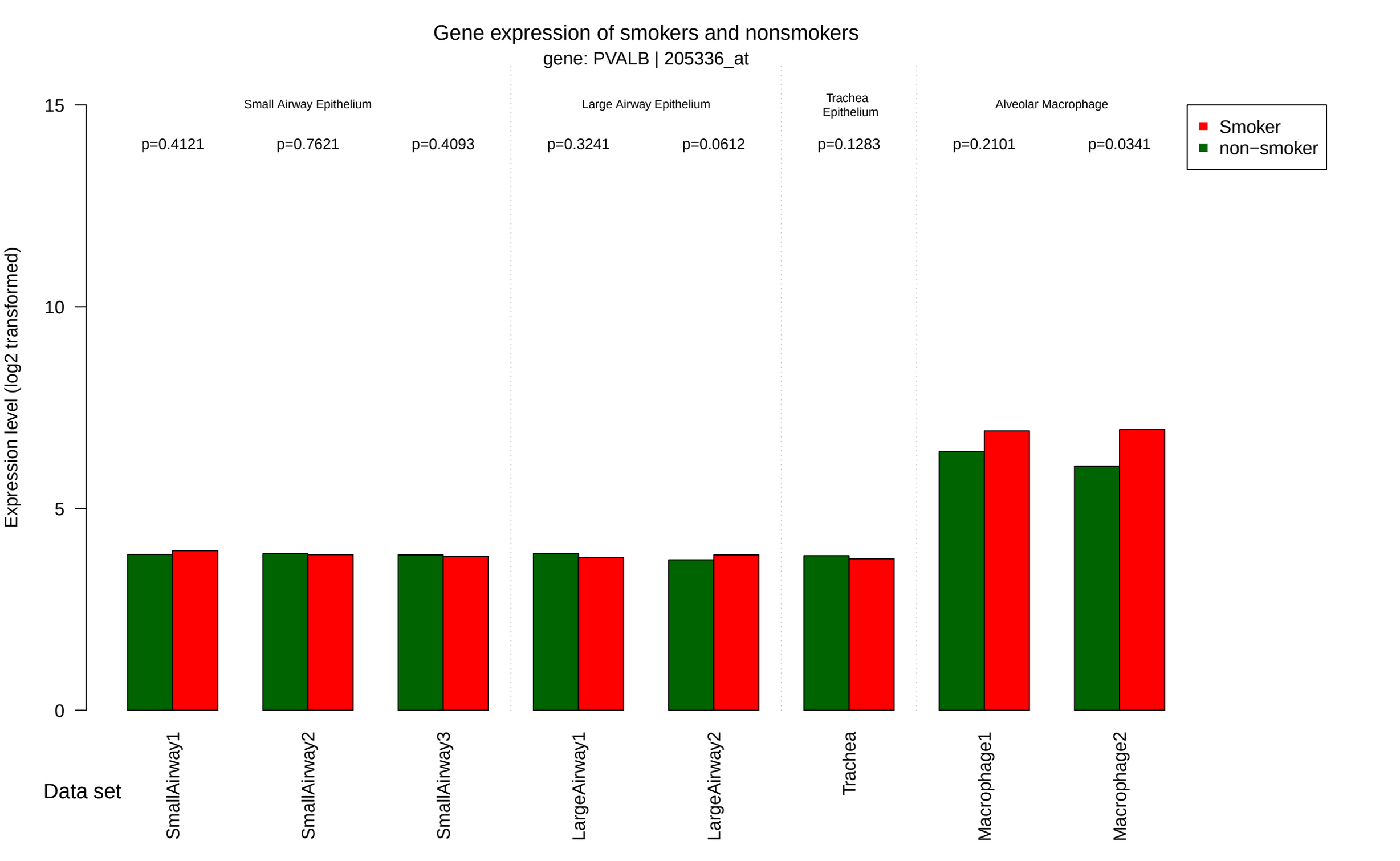
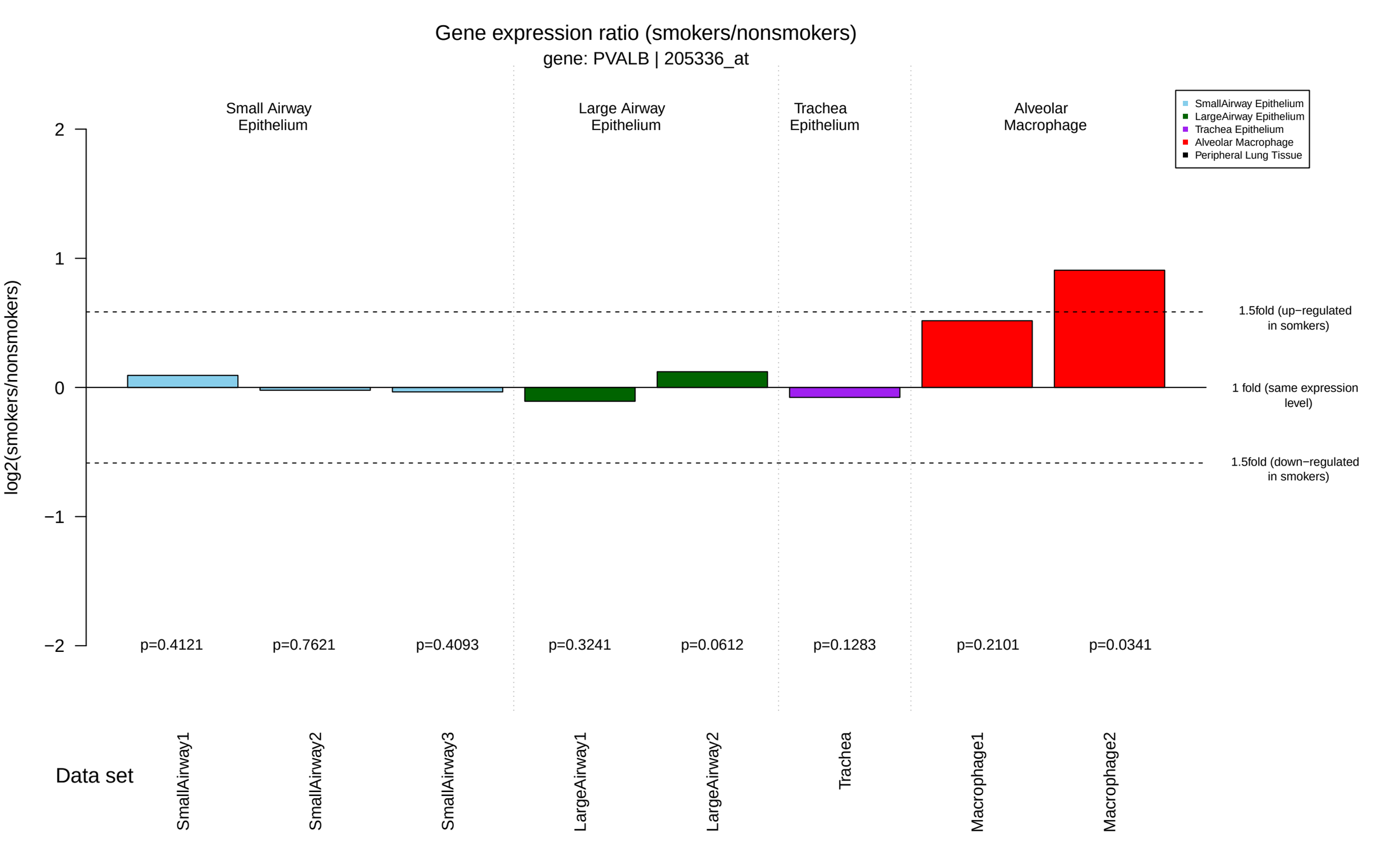
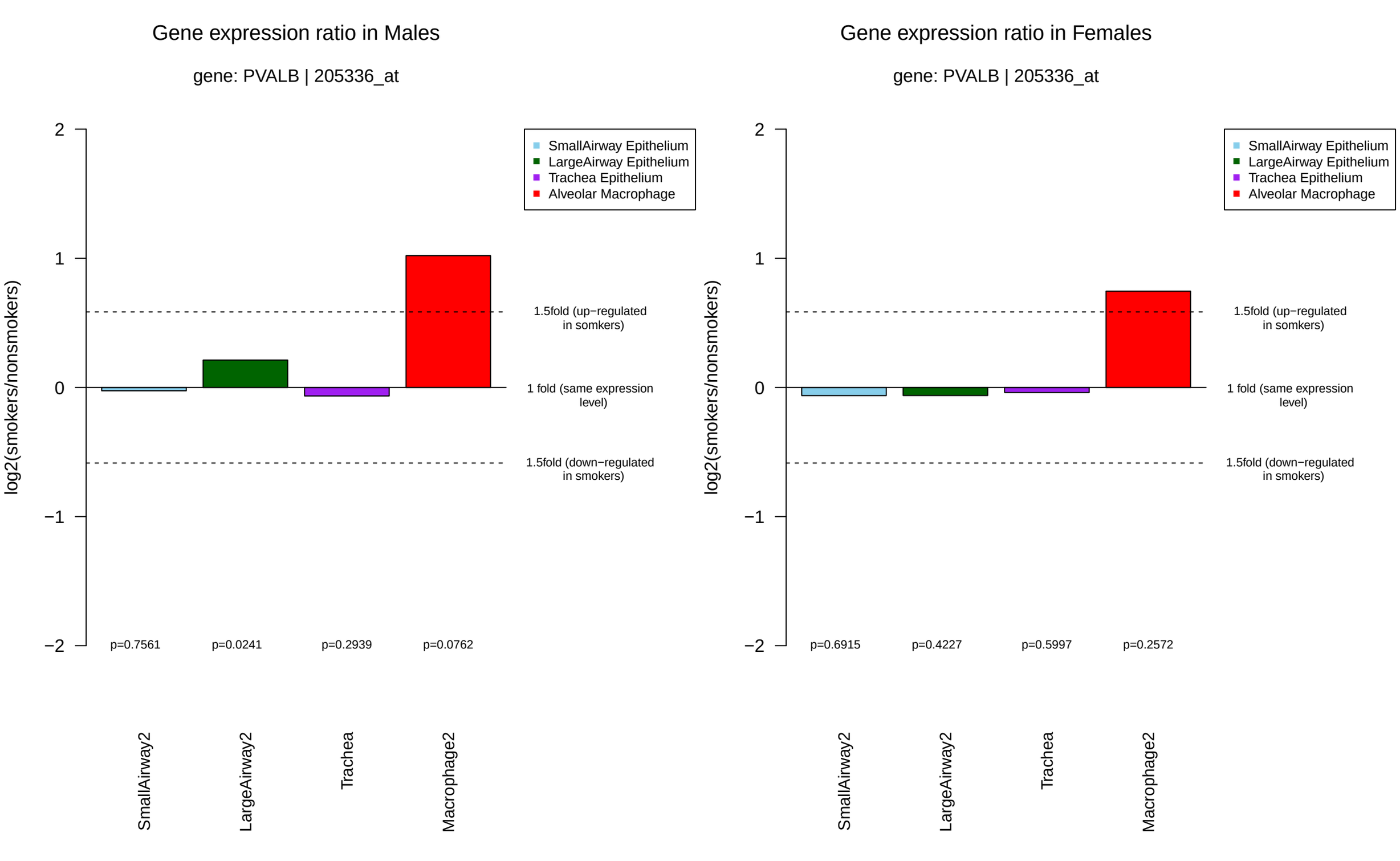
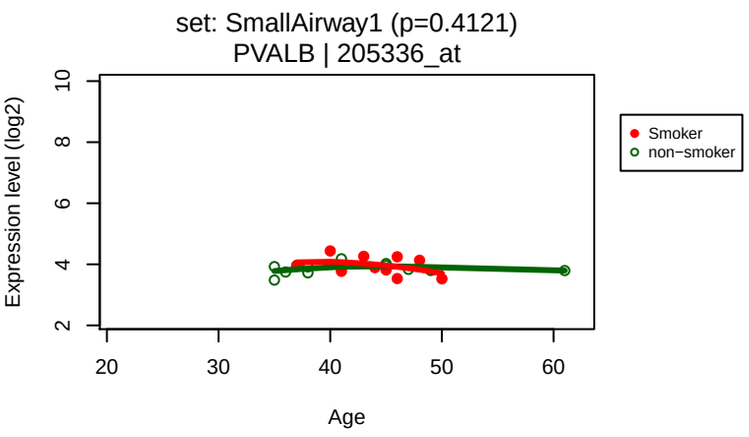
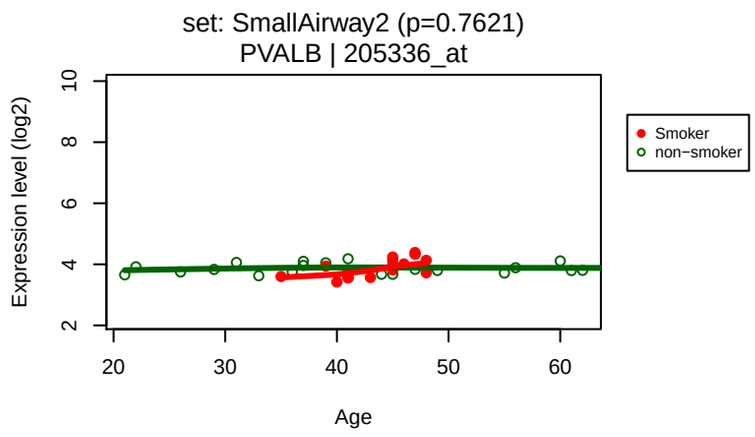
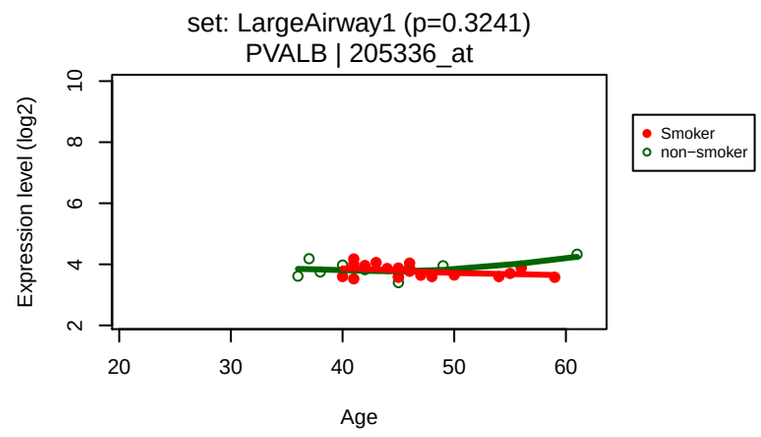

Gene expression of smokers and nonsmokers
gene: PVALB | 205336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4121
p=0.7621
p=0.4093
p=0.3241
p=0.0612
p=0.1283
p=0.2101
p=0.0341
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

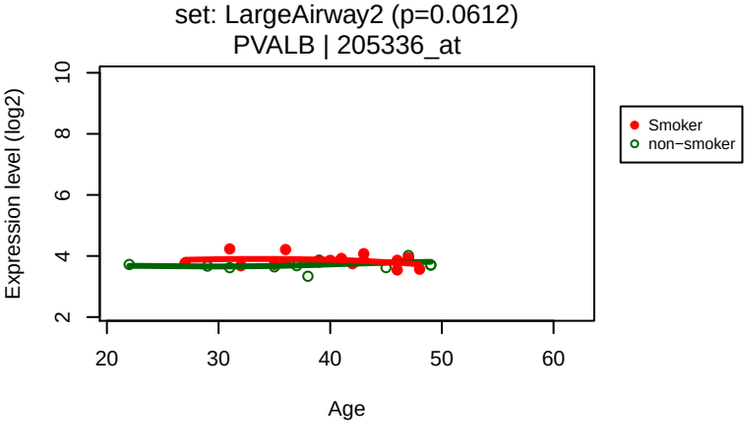

Gene expression ratio (smokers/nonsmokers)
gene: PVALB | 205336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4121
p=0.7621
p=0.4093
p=0.3241
p=0.0612
p=0.1283
p=0.2101
p=0.0341
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

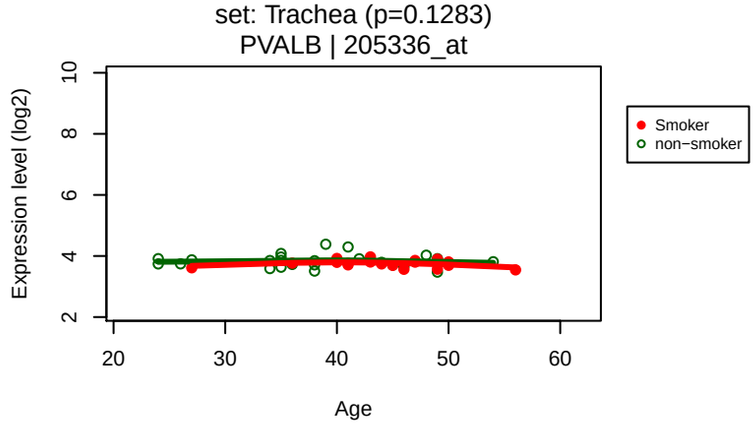

Gene expression ratio in Males
gene: PVALB | 205336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7561
p=0.0241
p=0.2939
p=0.0762
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PVALB | 205336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6915
p=0.4227
p=0.5997
p=0.2572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

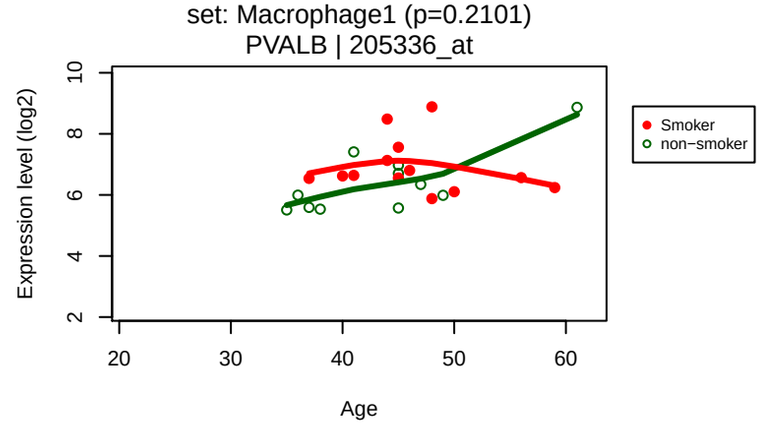

set: SmallAirway1 (p=0.4121)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker

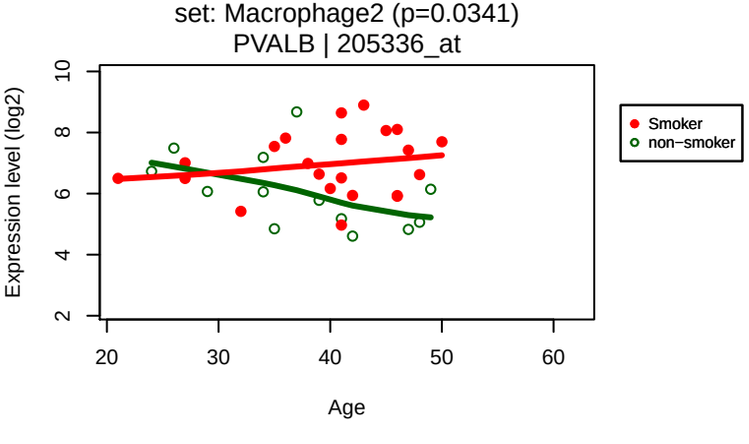

set: SmallAirway2 (p=0.7621)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker

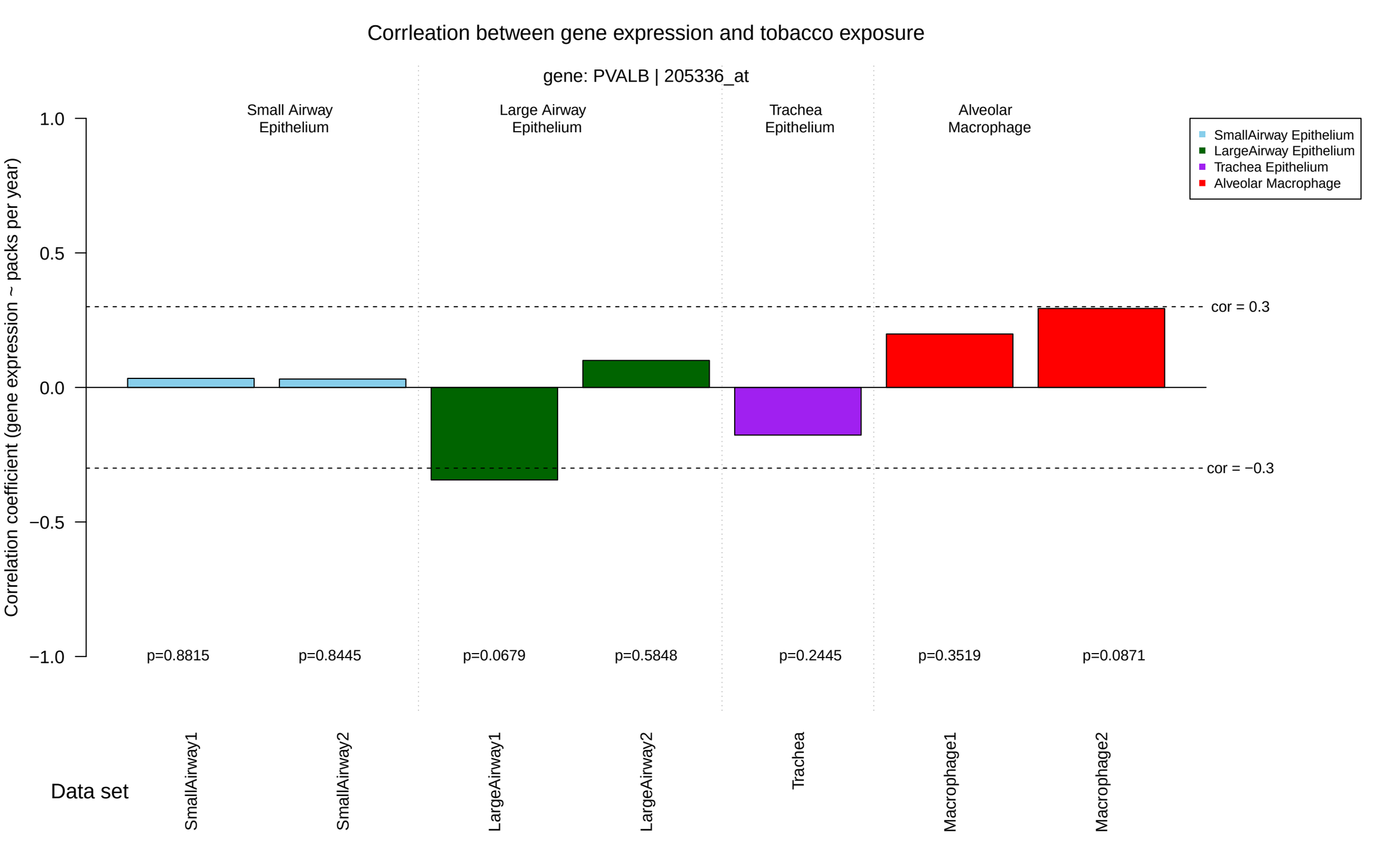

set: LargeAirway1 (p=0.3241)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0612)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1283)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2101)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0341)
PVALB | 205336_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PVALB | 205336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8815
p=0.8445
p=0.0679
p=0.5848
p=0.2445
p=0.3519
p=0.0871
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2